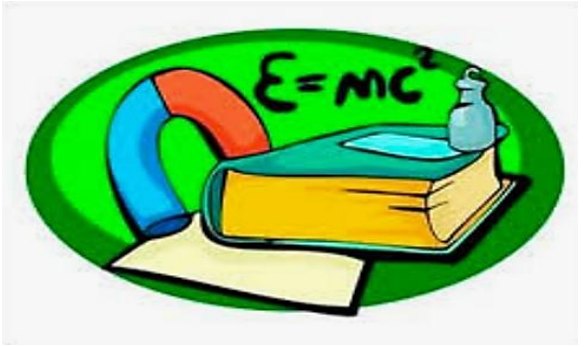
Fizika

9-sinf

Mavzu: **Masalalar yechish.**



Qattiq jismlarning mexanik xossalari

$$\Delta l = l - l_0 \qquad \varepsilon = \frac{\Delta l}{l_0} \qquad \varepsilon = \frac{\Delta l}{l_0} \cdot 100\%$$

$$\sigma = \frac{F}{S} \qquad \sigma = E \cdot |\varepsilon| \qquad F = \mathrm{ES}\frac{\Delta l}{l_0}$$

Test

Elastiklik (Yung) moduli qanday birlikda o‘lchanadi?

A) N/m B) $N \cdot m$ C) $Pa \cdot m$ D) Pa

To‘g‘ri javob: D) Pa

21-mashq 3-masala

Mustahkamlik chegarasi 0,5 MPa va zichligi 4000 kg/m^3 bo‘lgan gisht devorning balandligi eng ko‘pi bilan qancha bo‘lishi mumkin?

Berilgan:

$\sigma = 0{,}5\ MPa = 5 \cdot 10^5\ Pa$

$\rho = 4000\ kg/m^3$

$g = 9{,}8\ m/s^2$

Topish kerak: h−?

Formula:

$$\sigma = \frac{F}{S} \qquad F = mg$$

$$m = \rho V \qquad V = Sh$$

$$\sigma = \frac{mg}{S} = \frac{\rho Shg}{S} = \rho hg$$

$$h = \frac{\sigma}{\rho g}$$

Yechish: $h = \dfrac{5 \cdot 10^5 Pa}{4000 \frac{kg}{m^3} \cdot 9{,}8 m/s^2} = 12{,}7$ m

Javob: $h = 12,7\ m$

21-mashq 5-masala

Po‘latdan yasalgan sterjen uchiga 7,85 kN kuch qo‘yilganda u uzilib ketdi. Uning diametri qanday bo‘lgan?

Po‘lat uchun mustahkamlik chegarasi 170 MPa.

Berilgan:

$F = 7{,}85\ kN = 7850\ N$

$\sigma = 170\ MPa = 170 \cdot 10^6 Pa$

Topish kerak: S–?

Formula:

$$\sigma = \frac{F}{S}$$

$$S = \frac{F}{\sigma} \qquad S = \frac{\pi d^2}{4}$$

$$\frac{\pi d^2}{4} = \frac{F}{\sigma}$$

$$d^2 = \frac{4F}{\pi\sigma}$$

$$d = \sqrt{\frac{4F}{\pi\sigma}}$$

Yechish: $d = \sqrt{\frac{4 \cdot 7850\ N}{3{,}14 \cdot 170 \cdot 10^6 Pa}} = 7{,}7 \cdot 10^{-3}\text{m} = 7{,}7\ \text{mm}$

Javob: $d = 7{,}7\ mm$

21-mashq 6-mashq

Bir uchidan osib qo‘yilgan po‘lat sim suvga tushirilmoqda. Sim o‘zining og‘irligi ta’sirida uzilib ketmasligi uchun simning uzunligi qancha bo‘lishi kerak? Po‘lat uchun mustahkamlik chegarasi 170 MPa, zichligi $7800\ kg/m^3$ ga teng.

Berilgan:

$\sigma = 170\ MPa = 170 \cdot 10^6\ Pa$

$\rho_p = 7800\ \text{kg}/m^3$

$\rho_s = 1000\ \text{kg}/m^3$

Topish kerak: $l - ?$

Formula:

$$\sigma = \frac{F}{S} \qquad F = F_{og'} - F_A$$

$$F = mg - \rho_s V_j g$$

$$m = \rho_p V_j = \rho_p Sl$$

$$F = \rho_p Slg - \rho_s Slg =$$

$$= Slg(\rho_p - \rho_s)$$

$$\sigma = \frac{Slg(\rho_p - \rho_s)}{S}$$

$$l = \frac{\sigma}{g(\rho_p - \rho_s)}$$

Formulasi: $$l = \frac{\sigma}{g(\rho_p - \rho_s)}$$

Yechish: $$l = \frac{170 \cdot 10^6 Pa}{9{,}8\ m/s^2 \cdot (7800 \frac{kg}{m^3} - \frac{1000 kg}{m^3})} = 2551\ \text{m}$$

Javob: $\boldsymbol{l = 2551\ m}$

masala

Uzunligi 1,2 m va ko‘ndalang kesim yuzi 1,5 mm^2 bo‘lgan simga qanday kuch qo‘yilganda u osilganda sim 2 mm ga uzayadi.

Shu sim uchun Yung moduli 180 Gpa.

Berilgan:

$l_0 = 1{,}2\ m$

$S = 1{,}5\ mm^2 = 1{,}5 \cdot 10^{-6} m^2$

$\Delta l = 2\ mm = 2 \cdot 10^{-3}\text{m}$

$\sigma = 180\ GPa = 180 \cdot 10^{9}\text{Pa}$

Topish kerak: F—?

Formula:

$\sigma = \frac{F}{S}$; $\sigma = E|\varepsilon| = \text{E}\left|\frac{\Delta l}{l_0}\right|$

$\frac{F}{S} = \text{E}\left|\frac{\Delta l}{l_0}\right|$

$F = ES\left|\frac{\Delta l}{l_0}\right|$

Yechish: $F = 180{\cdot}10^{9}\text{Pa} \cdot 1{,}5 \cdot 10^{-6}\ m^2 \cdot \frac{2{\cdot}10^{-3} m}{1{,}2\ m} = 450\ \text{N}$

Javob: $\boldsymbol{F = 450\ N}$

Mustaqil bajarish uchun topshiriqlar

1. 21-mashqning 4-masalasini yechish.
2. 127-betdagi 8, 10- testlarni yechish.